

Quantitative Study: Intrahour Price Boundary Dynamics (NQ1)

Abstract: An exhaustive 20-year analysis (2006-2026) of intrahour market architecture. This study quantifies the predictive power of the 'Quarter 1' (0-15m) window in establishing hourly High/Low boundaries, calculates micro-temporal 'Minute 0' momentum, and tracks range expansion dynamics through subsequent violation sequences.

I. Methodology & Data Infrastructure

1. Parameters

- **Resolution:** 1-Minute OHLC Intervals (Full Historical Depth)
- **Time-Series:** America/New_York (EST/EDT) aligned.
- **Discretization:** Hourly sessions (60m) segmented into 15m Quarters (Q1-Q4).

2. Primary Metrics

- **Boundary Certainty:** Probability that the first 15 minutes (Q1) contains either the High or Low of the entire hour.
- **Micro-Certainty (m0):** Probability that the **very first minute** of a quarter establishes its 15-minute boundary.
- **Violation Dynamics:** The frequency and timing of Quarter boundaries being surpassed by subsequent price action.

II. Statistical Results

1. Q1 High/Low Predictive Probability

Shows how often the first 15 minutes holds the hourly boundary.

Hour	Total	Q1 as High	Q1 as Low	Q1 H or L (Influence)
18:00	5041	2858 (56.7%)	2919 (57.9%)	4635 (91.9%)
19:00	5033	1923 (38.2%)	2101 (41.7%)	3739 (74.3%)
20:00	5004	2073 (41.4%)	2184 (43.6%)	3918 (78.3%)
21:00	5004	1849 (37.0%)	2025 (40.5%)	3621 (72.4%)
22:00	5004	2033 (40.6%)	2120 (42.4%)	3820 (76.3%)
23:00	5008	1934 (38.6%)	2126 (42.5%)	3742 (74.7%)
00:00	5055	1884 (37.3%)	2013 (39.8%)	3643 (72.1%)
01:00	5131	1864 (36.3%)	2076 (40.5%)	3701 (72.1%)
02:00	5132	2071 (40.4%)	2343 (45.7%)	4076 (79.4%)
03:00	5132	2121 (41.3%)	2120 (41.3%)	4003 (78.0%)
04:00	5132	2114 (41.2%)	2254 (43.9%)	4070 (79.3%)
05:00	5141	1893 (36.8%)	2069 (40.2%)	3755 (73.0%)
06:00	5145	1949 (37.9%)	2061 (40.1%)	3789 (73.6%)
07:00	5147	1877 (36.5%)	2141 (41.6%)	3845 (74.7%)
08:00	5154	1620 (31.4%)	1630 (31.6%)	3164 (61.4%)
09:00	5160	865 (16.8%)	820 (15.9%)	1667 (32.3%)
10:00	5150	2019 (39.2%)	2265 (44.0%)	4093 (79.5%)
11:00	5152	1814 (35.2%)	2162 (42.0%)	3853 (74.8%)
12:00	5102	1820 (35.7%)	2070 (40.6%)	3740 (73.3%)
13:00	5026	1782 (35.5%)	1994 (39.7%)	3618 (72.0%)
14:00	4982	1682 (33.8%)	2031 (40.8%)	3593 (72.1%)
15:00	4980	1507 (30.3%)	1694 (34.0%)	3129 (62.8%)
16:00	4980	3370 (67.7%)	3646 (73.2%)	4848 (97.3%)

2. Micro-Temporal Precision (Minute 0)

Probability that the boundary (H or L) for a 15m quarter is formed in its **first minute**.

Hour	Q1 (H L)	Q2 (H L)	Q3 (H L)	Q4 (H L)
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18:00	45.2 43.5%	21.4 21.2%	23.7 22.3%	20.3 20.5%
19:00	23.9 23.6%	19.6 21.0%	21.2 23.0%	20.6 21.4%
20:00	27.0 24.4%	22.4 23.0%	21.9 21.0%	20.9 22.2%
21:00	22.6 22.8%	21.8 23.0%	22.4 22.1%	20.5 21.9%
22:00	23.0 21.6%	19.2 20.0%	21.5 23.0%	19.4 21.1%
23:00	21.0 22.3%	20.0 20.2%	23.3 22.8%	19.2 19.9%
00:00	21.6 21.4%	18.7 18.5%	22.9 22.8%	17.8 18.8%
01:00	23.4 23.5%	19.2 20.8%	20.8 22.0%	18.2 18.6%
02:00	27.9 29.4%	20.4 21.5%	22.2 23.8%	19.0 20.0%
03:00	25.1 23.9%	21.8 24.5%	22.4 23.0%	20.4 22.0%
04:00	23.9 26.1%	20.7 22.3%	21.6 22.7%	19.9 21.8%
05:00	22.7 24.0%	19.7 21.9%	21.3 23.0%	20.5 20.5%
06:00	23.6 22.4%	20.1 21.3%	21.2 22.4%	19.4 21.8%
07:00	23.7 27.7%	22.5 22.2%	22.6 23.0%	20.9 21.2%
08:00	23.9 24.4%	20.2 21.7%	31.0 31.4%	22.4 21.3%
09:00	23.9 24.5%	17.7 17.7%	22.9 24.3%	22.0 22.4%
10:00	23.8 23.8%	19.3 22.6%	21.2 22.2%	20.8 20.8%
11:00	20.5 24.0%	19.7 21.5%	22.0 23.7%	19.5 21.5%
12:00	22.2 22.8%	20.8 20.9%	21.0 21.9%	19.9 20.4%
13:00	21.9 21.6%	20.8 21.5%	21.6 21.4%	19.7 21.4%
14:00	22.9 22.5%	19.2 21.6%	19.2 24.2%	19.1 20.6%
15:00	22.8 21.9%	19.0 19.9%	20.4 23.2%	20.1 18.5%
16:00	31.7 40.8%	22.1 23.7%	34.8 40.0%	21.4 24.1%

3. Q1 Boundary Breakout Dynamics

Frequency of subsequent quarters (Q2, Q3, Q4) violating the initial Q1 High or Low.

Hour	Q1 High Violated in (Q2 Q3 Q4)	Q1 Low Violated in (Q2 Q3 Q4)
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18:00	27.4% 10.0% 5.7%	26.9% 9.1% 5.8%
19:00	39.4% 13.2% 8.9%	37.1% 12.8% 8.0%
20:00	39.3% 12.0% 6.8%	38.3% 11.8% 5.7%
21:00	43.3% 13.5% 5.8%	39.8% 12.8% 6.5%

22:00	39.9%	12.9%	5.7%	37.5%	12.6%	6.6%
23:00	41.1%	13.1%	6.1%	38.1%	12.3%	6.1%
00:00	41.0%	14.1%	6.8%	40.3%	12.7%	6.3%
01:00	40.8%	13.7%	8.2%	39.2%	11.9%	7.4%
02:00	39.4%	12.5%	7.6%	36.0%	11.1%	7.1%
03:00	42.6%	10.8%	5.2%	41.9%	11.1%	5.7%
04:00	42.1%	10.8%	5.8%	40.2%	10.3%	5.5%
05:00	44.3%	12.6%	6.0%	41.4%	11.4%	6.8%
06:00	42.6%	12.1%	7.3%	41.4%	12.1%	6.4%
07:00	44.4%	12.4%	6.7%	39.9%	11.6%	6.8%
08:00	47.3%	17.0%	4.1%	45.8%	17.6%	4.9%
09:00	47.8%	31.0%	4.3%	48.8%	31.4%	3.9%
10:00	44.4%	11.1%	5.3%	40.5%	9.6%	5.9%
11:00	48.2%	10.9%	5.7%	42.1%	10.6%	5.3%
12:00	45.5%	12.3%	6.5%	43.1%	10.3%	6.1%
13:00	45.9%	12.4%	6.2%	43.4%	11.5%	5.4%
14:00	47.4%	12.8%	6.1%	42.8%	10.8%	5.6%
15:00	47.4%	13.5%	8.9%	44.5%	12.3%	9.1%
16:00	6.9%	18.6%	6.6%	5.9%	16.3%	4.4%

4. Day of Week Variance

Comparison of 'Q1 Influence' (H or L established in Q1) across trading days.

Hour	Sun	Mon	Tue	Wed	Thu	Fri
18:00	97.0%	90.4%	91.3%	90.7%	90.8%	0.0%
19:00	75.7%	74.9%	74.3%	73.9%	72.8%	0.0%
20:00	77.1%	77.4%	79.5%	78.5%	78.9%	0.0%
21:00	71.7%	71.0%	72.8%	73.4%	72.9%	0.0%
22:00	76.0%	76.4%	76.3%	76.8%	76.3%	0.0%
23:00	74.2%	73.4%	75.9%	73.8%	76.4%	0.0%
00:00	0.0%	72.0%	72.6%	70.8%	71.2%	73.7%
01:00	0.0%	71.2%	74.9%	70.3%	72.0%	72.2%

Hour	Sun	Mon	Tue	Wed	Thu	Fri
02:00	0.0%	79.8%	80.5%	80.9%	77.7%	78.1%
03:00	0.0%	78.3%	77.3%	77.6%	77.0%	79.8%
04:00	0.0%	78.6%	79.6%	80.1%	79.1%	79.2%
05:00	0.0%	72.7%	76.1%	70.8%	74.2%	71.3%
06:00	0.0%	74.1%	74.0%	73.6%	72.9%	73.6%
07:00	0.0%	76.3%	74.8%	73.7%	72.9%	75.8%
08:00	0.0%	69.9%	66.0%	62.3%	55.1%	53.6%
09:00	0.0%	29.4%	31.0%	33.0%	31.6%	36.5%
10:00	0.0%	78.5%	78.5%	79.1%	79.7%	81.6%
11:00	0.0%	74.3%	71.3%	77.5%	75.4%	75.5%
12:00	0.0%	71.3%	73.6%	72.8%	75.2%	73.6%
13:00	0.0%	70.8%	71.8%	73.0%	72.2%	72.0%
14:00	0.0%	71.8%	68.0%	74.3%	74.4%	72.2%
15:00	0.0%	60.6%	60.2%	66.6%	65.0%	61.7%
16:00	0.0%	97.5%	96.5%	96.5%	97.3%	99.0%

III. Expert Tactical Analysis

- **High Influence Hour:** 16:00 possesses a 97.3% boundary formation in Q1. This hour favors early range establishment and subsequent rotation.
- **High Momentum / Trend Hour:** 09:00 shows a lower 32.3% Q1 influence, indicating a high probability of persistent expansion throughout the hour.